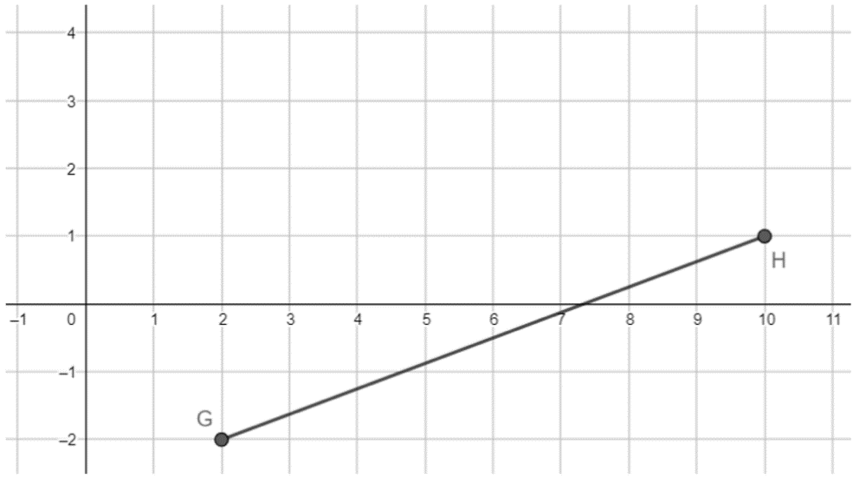
Geometry
Unit 13
Lesson 5 Homework

Name____________________
Date_____________________
Period___________________

Use the following information for questions 1-2. Point G has coordinates $(2, -2)$ and point H has coordinates $(10,1)$.



1. Complete the table to partition $\overline{GH}$ in a $3:1$ ratio.

	x**-value**	y**-value**
$\boldsymbol{G(2,-2)}$	$\frac{1}{4}$(____)	$\frac{1}{4}$(____)
$\boldsymbol{H(10,1)}$	$\frac{3}{4}$(____)	$\frac{3}{4}$(____)
Total		

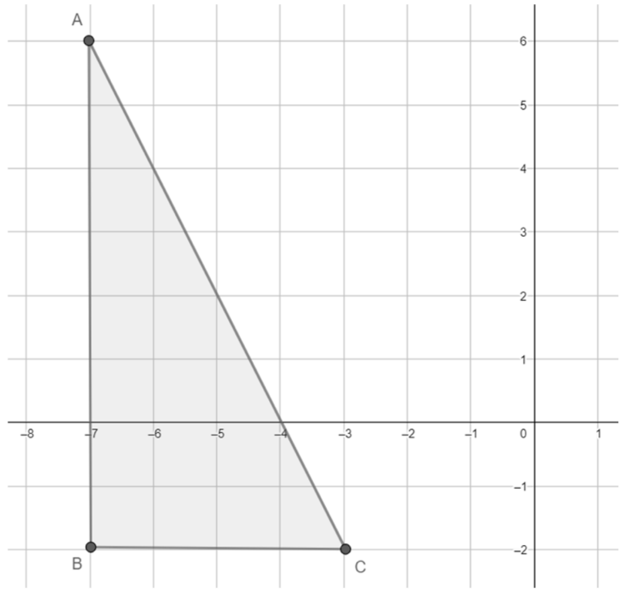
2. What is the midpoint of $\overline{GH}$?

Use the following information for questions 3-4. Line segment BT has endpoints at $B(-6,-4)$ and $T(4,6)$.

3. Write an expression for the point that partitions directed line segment BT in a $2:3$ ratio.

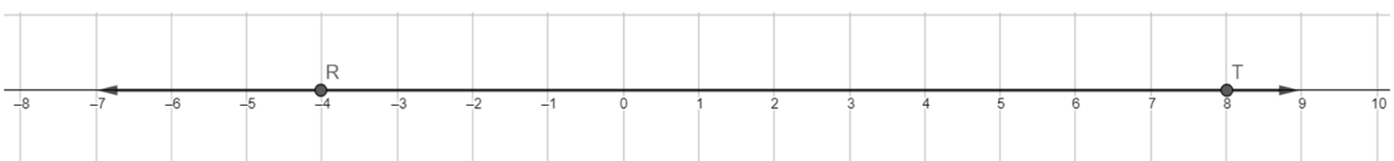
4. Use the expression to find the point that partitions directed line segment BT in a $2:3$ ratio.

Use the following information for question 5. Triangle ABC has vertices at $A(-7,6)$, $B(-7,-2)$, and $C(-3,-2)$.



5. Use segment partitioning to dilate triangle ABC using center B and scale factor $\frac{1}{2}$.

Use the following information for questions 6-7. A number line is shown. Each grid space is 1 unit.



6. Determine the point on the number line that satisfies the expression shown. Label the point G.

$$\frac{1}{4}(\text{point } R) + \frac{3}{4}(\text{point } T)$$

7. What is the ratio of $RG : GT$?